Soloeval at BAREC Shared Task 2026: QWK-Aligned Soft-Label Ordinal Classification with Auxiliary Surface Features for Arabic Readability Assessment

Turab Hussain Usmani
Independent Researcher
turabofficial2003@gmail.com

Abstract

We describe Soloeval's submission to the Sentence-Level Open Track of the BAREC Shared Task 2026 on fine-grained Arabic readability assessment. Rather than treating the 19-level readability scale as a scalar regression target, we fine-tune pretrained Arabic BERT encoders (MARBERTv2 and CAMeLBERT-Mix) with a soft-label cross-entropy objective whose target distribution is a Gaussian centered on the gold level, chosen to more directly reflect the Quadratic Weighted Kappa (QWK) evaluation metric's quadratic distance penalty. We combine this with lightweight auxiliary surface features (sentence length, word count, diacritic density) and F-fold cross-validation. Our best single-model configuration, CAMeLBERT-Mix trained with $F{=}3$ folds, achieves a QWK of 80.6 on the BAREC 2026 blind test leaderboard for the Open track. We report our full hyperparameter search, discuss the gap between our dev- and test-set results, and outline what we would try next.

1 Introduction

Automatic readability assessment predicts how difficult a text is for a reader, supporting applications from curriculum design to reading-level-aware content recommendation. For Arabic, this task is substantially harder than for English: Arabic is morphologically rich, frequently under-diacritized in real-world text, and spans a continuum from Modern Standard Arabic (MSA) to Classical Arabic and dialectal variation, all of which affect perceived difficulty independently of vocabulary or syntax. The BAREC Shared Task 2026 (Elmadani et al., 2026), built on the Balanced Arabic Readability Evaluation Corpus (Elmadani et al., 2025b), formalizes this as fine-grained 19-level ordinal classification at both the sentence and document level, evaluated primarily by Quadratic Weighted Kappa (QWK), a metric that penalizes predictions in proportion to the square of their distance from the true level.

This ordinal structure creates a specific modeling tension. Framing the task as scalar regression, as explored by several BAREC 2025 systems (e.g., Elmadani et al., 2025a; Trigui, 2025), optimizes a continuous mean-squared-error surface that has no explicit notion of the 19 discrete levels or the metric's quadratic penalty; framing it as plain 19-way classification with one-hot targets, conversely, treats a prediction of level 9 for a true label of 10 exactly as wrong as a prediction of level 19, which is precisely the behavior QWK is designed to punish more. In our own early experiments with a scalar-regression baseline, we observed validation QWK oscillate substantially across epochs even as training loss decreased monotonically – a symptom of this loss/metric mismatch rather than of insufficient training.

Our approach addresses this directly: we keep a 19-way classification head, but replace the one-hot cross-entropy target with a Gaussian distribution centered on the gold level (§3). This is a lightweight, architecture-agnostic alternative to more structurally involved ordinal regression frameworks such as CORAL (Cao et al., 2020), and is closely related in spirit to the differentiable soft-QWK objective explored concurrently by Trigui (2025) in the BAREC 2025 Strict track, though we apply it to a classification rather than a regression head. We combine this loss with a small set of auxiliary surface features and cross-validation across two pretrained Arabic encoders (MARBERTv2 and CAMeLBERT-Mix), evaluating what a single independent researcher, without ensembling multiple architectures at inference time or synthetic data augmentation, can achieve in the Open track under this design.

We participate only in the Sentence-Level Open Track, which permits any external resources. Our final submission reaches a blind test QWK of 80.6, and we discuss both what worked and what our results suggest we have not yet solved, particu-

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Sentence	Level
أرنب	1
كأن حدوج المالكية غدوةً خلايا سفينٍ بالنواصف من دد	19

Table 1: Example sentences at the lowest and highest BAREC readability levels.

larly the discrepancy between our development and blind test performance.

2 Background

2.1 Shared Task Description

BAREC-ST-2026 (Elmadani et al., 2026) is the second edition of the BAREC shared task, following BAREC 2025 (Elmadani et al., 2025a). It defines two subtasks – sentence-level and document-level readability assessment – each scored on the 19-level BAREC scale, where document-level readability is defined as the level of its most difficult sentence. Three tracks constrain what training data is permitted: the **Strict** track allows only the BAREC training set; the **Constrained** track additionally permits the SAMER Corpus (Alhafni et al., 2024) and SAMER Lexicon (Al Khalil et al., 2020); and the **Open** track, which we participate in, places no restriction on external resources. Annotation follows the guidelines of Habash et al. (2025). Table 1 shows two representative BAREC levels.

The primary evaluation metric is Quadratic Weighted Kappa (QWK), which penalizes predictions quadratically in their distance from the gold label. Secondary metrics are 19-level accuracy (Acc19) and its coarser collapses into 7, 5, and 3 levels (Acc7/5/3), adjacent (off-by-one) accuracy, and mean absolute distance (MAE/Dist).

2.2 Related Work

The first BAREC shared task (Elmadani et al., 2025a) drew 17 system description papers; the winning team, !MSA (Basem et al., 2025), reached 87.5 QWK sentence-level by confidence-weighted ensembling of four fine-tuned Arabic transformers (AraBERTv2, AraELECTRA, MARBERT, CAMeLBERT) with distinct loss functions per model, combined with synthetic data augmentation for rare levels. Several other BAREC 2025 systems targeted the QWK/loss mismatch we describe in §1 directly: Trigui (2025) trained AraBERT-v2 with a differentiable soft-QWK loss for the Strict track sentence-level task; Abdou (2025) instead left the base model's loss unchanged and applied conformal prediction as a post-hoc decoding step, using softmax-renormalized weighted averages over prediction sets to reduce high-penalty misclassifications, reporting consistent 1–3 point QWK gains across base models. Our approach is closer to Trigui's in modifying the training objective itself, but applies the QWK-alignment idea to a classification head via soft labeling rather than to a regression head via a differentiable kappa loss.

For pretrained encoders, we use MARBERTv2, from the ARBERT/MARBERT family (Abdul-Mageed et al., 2021), trained on a mix of MSA and dialectal Arabic tweets, and CAMeLBERT-Mix (Inoue et al., 2021), trained on a mixture of MSA, dialectal, and Classical Arabic text – a variant mixture that seemed a plausible fit for BAREC's span from simple MSA sentences to Classical Arabic poetry (Table 1).

3 System Overview

We frame sentence-level readability assessment as 19-way ordinal classification. Given a sentence, a pretrained encoder produces a [CLS] representation, which we concatenate with five auxiliary surface features before a two-layer classification head over the 19 levels.

Soft-label loss. Instead of a one-hot target, we construct a discrete Gaussian distribution over the 19 classes centered on the gold level y:

$$p(k \mid y) = \frac{\exp\left(-\frac{(k-y)^2}{2\sigma^2}\right)}{\sum_{j=1}^{19} \exp\left(-\frac{(j-y)^2}{2\sigma^2}\right)} \quad (1)$$

and minimize the cross-entropy between this target distribution and the model's predicted class distribution $\hat{p}$:

$$\mathcal{L} = -\sum_{k=1}^{19} p(k \mid y) \log \hat{p}(k) \quad (2)$$

where $y \in \{1, \ldots, 19\}$ denotes the integer gold readability level, $k \in \{1, \ldots, 19\}$ indexes the target readability classes, and σ is a dispersion parameter that controls the degree of label smoothing around the true level (fixed to $\sigma = 1.0$). In Equation (2), $\hat{p}(k)$ represents the softmax probability assigned by the classification head to class k.

We did not sweep σ exhaustively and note this as a natural next step (§5).

Auxiliary features. We concatenate five handcrafted features with the pooled [CLS] vector: normalized character count, word count, average word length, diacritic density (fraction of characters that are Arabic diacritics), and punctuation density. These are cheap to compute and give the model direct access to surface-level length and vocalization signals that plausibly correlate with BAREC level, without requiring the encoder to learn them purely from subword co-occurrence on a comparatively small fine-tuning set.

Cross-validation. For each encoder, we train with stratified F-fold cross-validation (evaluating $F \in \{1, 2, 3\}$; see (§4), which provides a stable estimate of validation performance and allows us to average probability predictions across fold models during inference.

4 Experimental Setup

Data. We use the official BAREC-ST-2026 sentence-level train, dev, and test splits, plus the official blind test set for final leaderboard submission, all supplied by the shared task organizers. Although the Open Track permits the use of external resources, we deliberately used no additional external text for training beyond these official splits; the auxiliary surface features are computed directly from the input sentence and introduce no outside data.

Preprocessing. Sentences are tokenized with each encoder's native tokenizer, truncated/padded to a maximum length of 128 subword tokens. Auxiliary features are computed on the raw sentence text prior to tokenization.

Models. We independently fine-tune MARBERTv2 (Abdul-Mageed et al., 2021) and CAMeLBERT-Mix (Inoue et al., 2021), evaluating each separately rather than ensembling their predictions.

Hyperparameters. We fine-tune with AdamW, a base learning rate of $2e{-}5$, a cosine learning-rate schedule with 10% linear warmup, batch size 16 with gradient accumulation of 2 (effective batch size 32), and mixed-precision (fp16) training. We searched over the number of cross-validation folds ($F \in \{1, 2, 3\}$) and the number of training epochs per fold ($E \in \{1, \ldots, 5\}$), selecting configurations by QWK on held-out splits. The smoothing parameter $\sigma = 1.0$ was fixed across all runs.

Metric	Score
QWK	76.7
Acc19	39.7
Acc19 ± 1	66.1
Dist (MAE)	1.4
Acc7	55.9
Acc5	62.2
Acc3	71.6

Table 2: Dev leaderboard result: an early $F{=}2$, 3-epoch configuration from our hyperparameter search, prior to selecting our final architecture and fold count.

Metric	$F{=}1$, **E = 1**	$F{=}1$, **E = 5**	$F{=}3$, **E = 3**†
QWK	79.7	78.9	**80.6**
Acc19	45.9	49.5	49.1
Acc19 ± 1	71.7	71.9	**73.6**
Dist (MAE)	1.1	1.1	1.1
Acc7	62.0	63.5	**64.5**
Acc5	68.5	69.1	**70.0**
Acc3	73.8	74.8	**75.1**

Table 3: Test-set results (CAMeLBERT-Mix) across three configurations from our search. † = our final blind test leaderboard submission (Open track, sentence-level), matching the officially reported score of 80.6 QWK.

Tools. We use the Hugging Face `transformers` (Wolf et al., 2020) and `datasets` libraries for model fine-tuning and data loading, and `scikit-learn` for metric computation and stratified fold splitting.

Metrics. We report all official BAREC metrics: QWK, Acc19, Acc7/5/3, adjacent (off-by-one) accuracy, and mean absolute distance (Dist), as defined in §2.

5 Results

Table 2 reports our development-set leaderboard result. Table 3 reports three configurations evaluated against the official test scorer during our hyperparameter search, the last of which (CAMeLBERT-Mix, $F{=}3$, 3 epochs) is our final blind test leaderboard submission.

Our final blind test QWK of 80.6 is below the 84.9 test QWK reported for BAREC 2025 conformal-prediction post-processing in the Strict track (although the results are not directly comparable because they correspond to different task editions and tracks), as well as the BAREC 2025 Open-track winning ensemble's 87.5 sentence-level QWK (Basem et al., 2025), which combined four architectures, weighted training, and synthetic

augmentation for rare levels—resources our single-encoder-per-run setup did not use. Within our own search, increasing F from 1 to 3 folds gave a consistent QWK improvement (79.7–78.9 $\rightarrow$ 80.6), while increasing epochs alone from 1 to 5 at $F{=}1$ improved Acc19 substantially (45.9 $\rightarrow$ 49.5) without a corresponding QWK gain, suggesting more training epochs sharpened exact-level predictions without proportionally reducing large misses – consistent with our hypothesis in §3 that fold averaging, rather than epoch count alone, is what helps QWK specifically.

We also note a gap between our development leaderboard result (76.7 QWK, Table 2) and blind test result (80.6 QWK, Table 3): our dev submission used an earlier, less-tuned configuration ($F{=}2$, 3 epochs) than our final blind test submission, so this is better read as evidence of gains from later hyperparameter search than as train/test distribution shift, though we cannot fully rule out the latter without re-running the earlier configuration on the blind test set.

Error analysis. Analysis of the per-level confusion matrix and Mean Absolute Error (MAE) on the test set reveals three primary error dynamics:

- **Dense middle-level alignment (Levels 4–14):** The model performs strongest on high-support intermediate levels. Level 10 achieves the lowest error (MAE $= 0.78$, with 60.3% exact-match accuracy), while levels 4, 5, 7, 8, 9, and 14 all maintain an MAE ≤ 1.22. The overwhelming majority of misclassifications across these levels fall within ±1 or ±2 bins along the diagonal, directly reflecting the regularizing effect of the Gaussian soft-label objective ($\sigma = 1.0$).

- **High-end truncation and mode collapse (Levels 17–19):** The model exhibits severe under-prediction at the extreme upper tail. There are zero predictions for levels 18 and 19, and only two predictions across the entire test set for level 17. Consequently, true instances of levels 17–19 are systematically pulled downward into levels 14–16, yielding sharply elevated error rates (MAE $= 2.49$ for level 17; 3.65 for level 18; 5.06 for level 19). This is driven by low support in the training split compounded by Gaussian label smoothing drawing probability mass toward high-density adjacent modes.

- **Low-end regression to modal classes (Levels 1–3):** Sparser low-end levels also exhibit higher variance (MAE $= 2.50$ for level 1; 2.11 for level 2; 3.11 for level 3). Rather than symmetric diffusion, errors skew upward toward major training modes (e.g., 44 out of 141 level-3 instances are predicted as level 5, and 25 as level 12).

These findings confirm that while soft-label smoothing effectively optimizes QWK by bounding dispersion in high-density regions, severe label sparsity at the ordinal extremes induces a regression-to-the-mean effect that requires explicit class-reweighting or targeted data augmentation.

6 Conclusion

We presented Soloeval's submission to the BAREC 2026 Sentence-Level Open Track: a soft-label ordinal classification objective, designed to align training more directly with the Quadratic Weighted Kappa evaluation metric than either one-hot classification or scalar regression, combined with lightweight auxiliary surface features and F-fold cross-validation over two pretrained Arabic encoders. Our best configuration, CAMeLBERT-Mix with 3-fold cross-validation, reached a blind test QWK of 80.6. Comparing across our own hyperparameter search suggests that fold averaging contributed more to QWK specifically than additional training epochs did, which improved exact-match accuracy without a matching QWK gain – a distinction that would have been invisible had we optimized only for Acc19. We did not sweep the soft-label variance σ, did not ensemble across architectures at inference time (only per-fold within a single architecture), and did not use any Open-track-permitted external data or augmentation beyond the official BAREC splits; each of these is a natural next step, and the gap to the strongest reported BAREC 2025 Open-track result (Basem et al., 2025) suggests cross-architecture ensembling and rare-level augmentation in particular would likely be the highest-leverage additions.

Limitations

Our study has several limitations:

- **Extreme-level sparsity and mode collapse:** The Gaussian soft-label objective exacerbates under-prediction at the ordinal extremes (levels 1–3 and 17–19). Because support is sparse

at the tail ends of the training set, symmetric label diffusion pulls probability mass toward high-density adjacent modes, leading to zero predictions for levels 18 and 19.

- **Fixed smoothing scale:** We set $\sigma = 1.0$ a priori to account for adjacent off-by-one errors. We did not conduct a systematic hyperparameter sweep over σ or explore class-dependent adaptive variances, which could mitigate regression-to-the-mean on rare classes.
- **Absence of cross-architecture ensembling:** We trained and averaged folds within individual architectures rather than ensembling heterogeneous model families (e.g., combining MARBERT and CAMELBERT).
- **No external data utilization:** Despite participating in the Open Track, we did not leverage outside corpora or synthetic text simplification datasets for data augmentation, leaving the performance headroom of external data unexplored.

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